Abstract citation ID: luae218.010

P-08 OLD ETIOLOGY, NEW CHALLENGES: MANAGING WATERHOUSE FRIDERICHSEN SYNDROME IN A PEDIATRIC PATIENT WITH SIGNIFICANT COMORBIDITIES

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Introduction: Waterhouse-Friderichsen syndrome (WFS) is a rare but life-threatening condition characterized by bilateral adrenal hemorrhage, typically precipitated by severe systemic infection. This syndrome can lead to chronic primary adrenal insufficiency, a condition that poses significant management challenges, particularly when accompanied by major comorbidities. **Clinical Case:** We report the case of a young boy who developed Neisseria meningitidis type B meningitis at 3 months of age, followed by purpura fulminans, septic shock and coma, necessitating orotracheal intubation and mechanical ventilation. Acute phase management included fluid resuscitation, antibiotics, osmotic diuretics, immunoglobulins, antipyretics and plasma. During hospitalization, the development of progressive generalized skin hyperpigmentation raised the suspicion of adrenal insufficiency, confirmed by cortisol (4.60 µg/dL) and ACTH (330 pg/mL) levels. Glucocorticoid replacement therapy was initiated, with a favorable recovery and no neurological deficits at discharge. Several months later, the patient began experiencing recurrent seizures, subsequently diagnosed as focal generalized epilepsy. Treatment with valproic acid was initiated, with the dosage gradually increased due to inadequate seizure control and lack of responsiveness to initial therapy. Despite appropriate glucocorticoid replacement therapy, the patient experiences nausea, vomiting, and apathy for several hours following each epileptic episode. Further complicating the clinical picture, the patient associates generalized epi-metaphyseal dysplasia, limb-length discrepancy and other skeletal deformities (Figure). Given these clinical features, genetic testing was performed using a panel of genes associated with skeletal dysplasia. This revealed a heterozygous variant in the TCIRG1 gene (c.907+4A>T), associated with autosomal recessive osteopetrosis. However, since both alleles must be mutated for the condition to manifest, the patient is considered a healthy carrier, which does not account for the observed phenotype. Subsequent genetic testing for Beckwith-Wiedemann syndrome revealed no pathological findings, leaving the etiology of the skeletal dysplasia currently unexplained, highlighting the complexity of the case. **Conclusion:** In 21st century, when most cases of adrenal insufficiency are of autoimmune etiology, we are facing a rare cause of adrenal insufficiency with potentially fatal prognosis in a 3 month-old infant. While the therapeutic approach of WFS is difficult, the added neuropsychiatric complications further challenges the management of chronic adrenal insufficiency, especially in the absence of established guidelines for glucocorticoid dose adjustments in epileptic patients. Complex cases require enhanced interdisciplinary collaboration and further research to develop evidence-based guidelines for managing adrenal insufficiency with comorbid neurological conditions.



Figure 1: Generalized lesional hyperpigmentation
The image shows an infant patient recovering from an episode of meningitis, complicated by disseminated purpura and adrenal insufficiency. These conditions have caused darkening of the skin and hyperpigmentation of healing purpuric lesions.